Synthetic Cultural Agents from Aggregate Anchors

A Falsifiable Protocol for Constructing Population-Level Choice Policies

“Population-level” denotes the source of the anchor, not an estimated human response distribution.

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Status of this version. This is a methods-first working-paper draft. Its conceptual protocol, proposed rejection design, and scope boundary are stated as commitments. The multi-country experiment, placebo controls, and pipeline-sensitivity results are intentionally represented by *predeclared planned* result shells rather than invented findings; this draft is not a public preregistration. A verified USA/Mexico pilot is retained only as an Appendix diagnostic.

Abstract

Social scientists increasingly use language models to simulate survey respondents or to align models with named communities. Both approaches obscure a central question: what exactly enters the model, and what evidence would justify interpreting its output as population-relevant? We propose a constructive alternative. A researcher declares an *aggregate anchor*—a vector of population-level measurements—and freezes a protocol that maps the anchor into synthetic pairwise data and then into a language-model choice policy. Country or identity labels are absent from primary inference prompts. The resulting object is therefore not an imitation of an identified person or group, nor an estimated population response distribution; it is an aggregate-anchor-induced synthetic choice policy. Its scientific value is empirical, not assumed. We specify five testable links: anchor transmission, policy encoding, anchor relevance relative to permuted-anchor placebos, external transport to independent human survey distributions, and sensitivity to upstream generation choices. We implement the protocol with country-level Global Preference Survey anchors, synthetic preference pairs, and Direct Preference Optimization, and state a planned World Values Survey analysis with non-LLM, base-model, cultural-prompt, and placebo baselines. The paper contributes a transparent way to construct and potentially reject synthetic cultural policies while drawing a sharp boundary around claims it cannot support: individual representation, full preference-distribution recovery, and downstream causal or structural inference.

Keywords: synthetic cultural agents; aggregate anchors; language models; preference optimization; survey simulation; falsification; culture

1. A construction problem, not an impersonation problem

Language models are now routinely asked to stand in for people: a prompt supplies a demographic profile, a country name, or a social identity, and the model is asked to answer a survey “as” that person or population. This practice is attractive because it is cheap and scalable. It is also scientifically precarious. Prompted synthetic respondents can be highly sensitive to wording and model version, and their distributions can differ materially from human survey distributions [1, 2]. More fundamentally, an inference-time persona makes it difficult to distinguish information supplied by the researcher from stereotypes or associations retrieved by the model.

The alternative is not to claim that a model has discovered a culture. It is to construct a model object from declared population-level information and then submit that construction to tests that can fail. We call the resulting object an *aggregate-anchor-induced synthetic choice policy*. The policy is built from a country-level preference profile, but it is not asserted to represent any individual in that country. This boundary matters: evidence that models can flatten or misportray identity groups is a warning against treating synthetic outputs as voices of the people they purport to emulate [7].

Our contribution is a protocol, not a theorem of human-preference recovery. A protocol specifies: which aggregate measurements enter; how they are rendered; how scenarios, response pairs, labels, and quality filters are generated; how the policy is estimated; and which internal, placebo, external, and sensitivity tests decide whether the construction is useful. The protocol is constructive because it creates a policy under researcher-authored rules. It is falsifiable because each link in the construction makes a separate empirical commitment.

The first implementation uses the Global Preferences Survey (GPS), which reports country-level profiles for trust, risk taking, patience, altruism, and positive and negative reciprocity [4]. These anchors yield synthetic preference pairs; Direct Preference Optimization (DPO) fits lightweight country-specific adapters relative to a common base policy [6]. The planned primary external criterion is a theory-mapped battery of World Values Survey (WVS) item distributions. Crucially, country information is excluded from the primary evaluation prompt. This removes a direct country-token channel; it does not prove that all country-associated information is absent from base-model knowledge, scenario wording, or the profile. Thus the relevant question is not whether a model can retrieve a country label at inference. It is whether a policy induced by a declared aggregate anchor has externally useful behavior under a transparent, fixed procedure.

This paper makes four contributions. First, it defines the aggregate-anchor construction problem and its scientific object. Second, it provides a modular protocol with anti-leakage, provenance, and reproducibility requirements. Third, it makes the protocol empirically testable through five links: anchor transmission, policy encoding, anchor relevance, external transport, and operator robustness. Fourth, it supplies a predeclared multi-country design that contrasts the constructed policy with base-model, country-prompt, non-LLM, and permuted-anchor baselines. The results of that design are deliberately not pre-judged here.

2. Positioning: where this protocol differs

The closest literatures differ in both their information source and the location at which that information enters the model. Early "silicon sample" work conditions models on detailed respondent profiles and asks whether generated responses resemble human samples [1]. Subsequent evidence finds instability, distributional mismatch, and inferential risks in such replacement exercises [2]. A second family fine-tunes models on human preference or survey-response data. These approaches may be appropriate when individual records or empirical response distributions are available; their claim is correspondingly closer to learned human response behavior. A third family evaluates or aligns models against named cultural communities.

Table 1: Four approaches to LLM behavior associated with social-science targets. The final row is this paper's object.

Approach	**Training information**	**Conditioning locus**	**Central claim**
Persona prompting	identity or demographic profile	inference prompt	model imitates a named respondent/group
Microdata fine-tuning	individual responses or distributions	weights and/or prompt	model learns observed response behavior
Cultural alignment	named community preferences or norms	weights and/or prompt	model aligns to an identified target
Aggregate-anchor SCA	synthetic pairs induced by declared aggregate measurements	weights; no country prompt in primary evaluation	a protocol-induced policy has passed stated tests of transport and robustness

The protocol uses familiar computational components but makes no novelty claim for them. A language model provides conditional probabilities over candidate responses; DPO provides a tractable way to fit a policy from pairwise preferences; low-rank adaptation makes separate adapters practical [6, 5, 3]. Our contribution is to specify how aggregate measurements enter the synthetic-data operator and how the resulting policy is subjected to tests that distinguish construction success from merely plausible prose.

3. The scientific object and its boundary

3.1 Aggregate anchors

For country c, let

$$z_c = (z_{c1}, \ldots, z_{cK}) \in \mathbb{R}^K \quad (1)$$

be a vector of pre-existing population-level measurements selected before training. In the first implementation, $K = 6$ and z_c contains GPS country scores, represented in the pipeline as standardized deviations from global means. An aggregate anchor is an input to a construction procedure. It is *not* an individual record, a country name shown at inference, or a moment condition automatically satisfied by a trained policy.

3.2 A frozen construction protocol

We define a protocol as

$$\mathcal{P} = (Z, R, G, S, Q, A, E), \tag{2}$$

where Z records the anchor source and country frame; R is the deterministic anchor-to-profile rendering rule; G is the scenario, response-pair, and label-generation procedure; S is scoring, quality control, and filtering; Q records prompts and direct-token anti-leakage restrictions; A is the policy estimator and its training configuration; and E is the planned evaluation, placebo, and sensitivity design. Conditional on an archived run manifest, the model-mediated stages of G and S are stochastic realizations; the protocol is not equivalent to a deterministic mapping from z_c to data.

Given a reference policy π_{ref}, the protocol generates a country-specific synthetic dataset $\mathcal{D}_c = \mathcal{G}_{\mathcal{P}}(z_c)$ and fits

$$\widehat{\pi}_c^{\mathcal{P}} = \mathcal{A}(\mathcal{D}_c; \pi_{\text{ref}}) \,. \tag{3}$$

Equation (3) defines a conditional distribution over candidate responses induced by a declared anchor and a frozen protocol. It does *not* assert $\widehat{\pi}_c^{\mathcal{P}} = P_{\text{human},c}$, where $P_{\text{human},c}$ denotes a human response distribution. That equality is an empirical transport hypothesis, not a definitional fact.

3.3 What this construction does not identify

Country averages do not identify which individuals hold which preferences, how preferences co-vary with demographic characteristics, or the full within-country distribution. The protocol therefore does not claim to recover individual heterogeneity, subgroup distributions, or a country's latent culture. Nor does it license reduced-form causal estimation, structural parameter recovery, or policy counterfactuals without a separate measurement and identification argument. These are not rhetorical caveats; they determine the object of study.

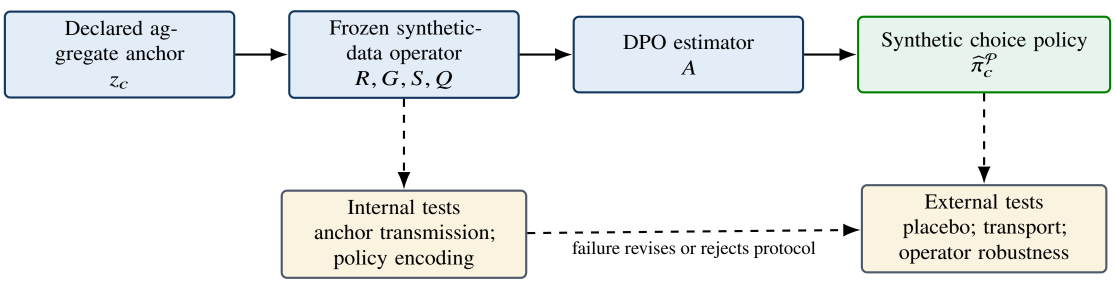

Figure 1: The paper's scientific object. Solid arrows construct a policy. Dashed arrows are tests, not guarantees. A failed test rejects a claim about this protocol; it does not prove that a population lacks the relevant preference.

4. From anchors to synthetic choice policies

4.1 Anti-leakage and profile rendering

The primary implementation begins with the GPS vector and renders it into an anonymized quantitative profile. The rendering states the magnitude and direction of each standardized score without inserting a country name. In the primary evaluation prompt, country and demographic labels are likewise omitted. This direct-token anti-leakage design does not establish cultural validity or prove that country-associated information is absent. It establishes a narrower, auditable fact: any difference between adapters cannot be attributed to country tokens in the primary inference prompt. The claim must be complemented by audits of scenario text, profile re-identifiability, and whether a classifier can infer country from generated artifacts.

4.2 Synthetic data operator

The operator has five conceptually distinct stages. First, a teacher model proposes behaviorally distinct facets and culturally neutral scenarios for each preference dimension. Second, a generator creates two plausible, opposing responses to each fixed scenario while being instructed to hold non-target traits constant. Third, an anchor-aware selector chooses the response aligned with the country profile. Fourth, an independent scorer assigns both responses scores on all six dimensions. Fifth, deterministic quality-control rules retain pairs only if the scorer reports adequate target separation and acceptable directional consistency.

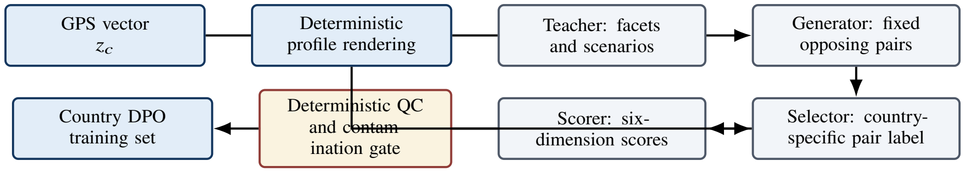

Figure 2: Current implementation architecture. The profile rendering is deterministic; the teacher, pair generator, selector, and scorer are separable model-mediated stages. The scientific-design appendix audits these choices and motivates ablations.

This decomposition matters. Calling every upstream choice "teacher bias" obscures the actual sources of dependence: scenario construction, response-pair generation, country-specific selection, scorer judgments, and filter thresholds may each change the synthetic training distribution. The empirical design therefore treats the full operator as an object of sensitivity analysis.

4.3 Policy estimation

For synthetic triplets $(x, y_w, y_\ell) \in \mathcal{D}_c$, DPO fits a policy relative to π_{ref} by minimizing

$$\mathcal{L}_{\text{DPO}}(\pi) = -\mathbb{E}_{(x,y_w,y_\ell)\sim\mathcal{D}_c}\left[\log\sigma\left(\beta\log\frac{\pi(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)}\right)\right]. \tag{4}$$

DPO is a pairwise-preference estimator with reference-policy regularization [6]. Here it fits synthetic preferences generated by $\mathcal{P}$. It is not a constrained estimator that mechanically enforces $\mathbb{E}[f(Y)] = z_c$, and we make no such claim.

5. Making the construction empirically rejectable

The phrase "falsifiable" is useful only if it names propositions, estimands, and failure consequences. This working-paper version supplies the test architecture, but not a public timestamped registration or final numerical thresholds; it should therefore be read as a proposed rejection design. Before any scaled outcome is inspected, the archived analysis plan will fix the unit of analysis, primary metric, comparison, practical margin, randomization/materialization count, uncertainty procedure, and country/domain/protocol decision rule for every row in Table 2. Each test concerns a distinct link; passing one does not imply passing another.

Table 2: The five rejection points of an aggregate-anchor construction.

Claim	Predeclared evidence	Interpretation of failure
Anchor transmission	label/sign agreement, target separation, QC rates, anchor swaps	the synthetic operator does not express the declared input reliably
Policy encoding	held-out synthetic-pair recovery; policy-distance stability	DPO does not retain the contrast present in the synthetic data
Anchor relevance	real-anchor versus permuted-anchor placebo adapters	apparent gains may reflect generic synthetic tuning rather than anchor information
External transport	independent human survey distributions versus base, prompt, and non-LLM baselines	this protocol does not generalize to the chosen external criterion
Operator robustness	scenario, generator, selector, scorer/QC, materialized-data, and training-seed ablations	the conclusion depends on arbitrary or unstable upstream choices

5.1 Internal tests: transmission and encoding

Anchor transmission asks whether changing z_c changes labels and retained pairs in the direction dictated by the declared rule. It is assessed with country-by-dimension label summaries, quality-control pass rates, target-versus-nontarget separation, contamination ratios, and anchor-swap diagnostics. The current implementation makes an especially important test available: because the selector is instructed to implement a sign rule, the primary sensitivity comparison must contrast a deterministic sign-labeling operator with the legacy LLM selector. Policy encoding asks whether the trained adapter reproduces held-out synthetic contrasts and whether independently trained policies are stable under repeated materializations of the same protocol.

5.2 Anchor relevance: a restricted permutation distribution

A central negative control draws many restricted permutations of country anchors before synthetic generation while retaining all other stages. The final plan will predeclare any strata used to preserve region, sample availability, or anchor magnitude, and will evaluate the real-anchor statistic in its resulting empirical null distribution. Each placebo uses the same data volume, model family, and training budget as the reference protocol. If real-anchor adapters do not outperform this distribution on predeclared external outcomes, the data do not support the claim that aggregate anchors add useful information beyond generic synthetic fine-tuning.

5.3 External transport and uncertainty

External transport is a predictive-criterion claim: the policy's induced option probabilities improve alignment with independently collected human response distributions. It is neither recovery nor validation of a complete country distribution. For a question q with valid options $\mathcal{Y}_q$, let $\ell_{\widehat{\pi}_c}(y \mid q)$ denote the total log probability assigned to the token sequence encoding answer option y under a fixed prompt and option-order rule. The policy yields

$$\widehat{P}_c(y \mid q) = \frac{\exp\{\ell_{\widehat{\pi}_c}(y \mid q)\}}{\sum_{j \in \mathcal{Y}_q} \exp\{\ell_{\widehat{\pi}_c}(j \mid q)\}}, \tag{5}$$

and the primary descriptive discrepancy is total variation distance,

$$\mathrm{TVD}(\widehat{P}_c, P_c) = \frac{1}{2} \sum_{y \in \mathcal{Y}_q} |\widehat{P}_c(y \mid q) - P_c(y \mid q)|. \tag{6}$$

For each country–item cell, P_c is the survey-weighted empirical option distribution after a frozen harmonization map for option wording, order, invalid outputs, and Likert categories. The analysis will report absolute TVD and paired differences from every baseline, together with practical margins. It will distinguish finite-survey sampling uncertainty, country and item aggregation, materialized synthetic datasets, and DPO training seeds; a hierarchical or randomization-based aggregation rule will be fixed before outcome inspection. A bootstrap over survey items is useful only as a robustness calculation for the declared item battery—not as respondent sampling uncertainty, an item-universe inference, or uncertainty about an upstream model-mediated pipeline.

6. Empirical implementation: predeclared design

6.1 Data roles and country frame

The first implementation uses GPS as the anchor source and WVS Wave 7 as the primary independent outcome surface. Before a scaled run, the authors will archive an immutable country inclusion/exclusion rule, a named 10–15-country panel from the GPS×WVS intersection, and a frozen theory-led item battery. The battery must separately list construct-linked confirmatory outcomes—with an expected direction or distributional relation for each GPS dimension—and descriptive/exploratory outcomes. The existing 35-item pilot map contains 30 mapped items and five demographics; it is not itself the confirmatory map for the scaled study.

AmericasBarometer and the Integrated Values Surveys may serve as secondary instrument or temporal replications if multi-country overlap is adequate. World Bank indicators are contextual covariates for exploratory heterogeneity analysis, not direct choice-distribution criteria.

6.2 Baselines and primary comparison

Every country is evaluated against four mandatory baselines: (i) the unmodified base model; (ii) a country-named cultural-prompt baseline; (iii) a named non-LLM aggregate benchmark fixed before outcome inspection; and (iv) the restricted permuted-anchor distribution. The country prompt is deliberately included despite violating direct-token anti-leakage because it answers a distinct practical question: whether an explicit persona retrieves useful information more effectively than the constructed policy. The analysis must report provenance-sensitive construction success separately from predictive alignment; a cultural prompt may be practically predictive while failing the former objective.

The primary cross-country claim will compare each fixed policy to each country's human distribution. Under unconditioned prompts, this is *fixed-policy proximity*, not evidence that the model conditions on a country at inference. Country-level permutation tests and cross-country patterns become informative only with more than two adapters. The design will report gains and failures, rather than defining success as a uniformly positive average.

6.3 Operator-sensitivity panel

A stratified 4–6-country subset is run before the full panel. The reference protocol is compared with alternative scenario banks, alternate pair-generation materializations, deterministic-sign versus legacy-selector labels, QC-threshold variations, a scorer-diagnostic-only/no-QC arm where feasible, restricted anchor permutations, and repeated DPO seeds. Each arm reports effects on (i) synthetic-data diagnostics, (ii) learned-policy stability, and (iii) external transport. At least three independently materialized scenario/pair banks are retained per condition. The full panel proceeds only if this subset establishes adequate anchor transmission, excludes obvious high-contamination pairs under a predeclared rule, and rejects the basic placebo explanation.

7. Planned reporting structure

Status. This section contains the planned reporting logic and table shells. It will be completed only from frozen artifacts produced under the protocol in Section 6. No scaled empirical outcome is claimed in this draft.

7.1 Anchor transmission and synthetic-data quality

Table 3 will report country-by-dimension label agreement, QC retention, target separation, and contamination. Failure is informative: a low pass rate, directional reversal, or high cross-dimension contamination is evidence against the operator, not a reason to selectively discard countries.

7.2 Policy encoding and repeatability

The next report will separate held-out synthetic-pair recovery from repeated-run policy stability. A policy that perfectly fits a single synthetic dataset but varies strongly across regenerated datasets has not demonstrated a stable aggregate-anchor construction.

7.3 External transport, placebo, and failure cases

The primary table will compare reference adapters, base models, cultural prompts, non-LLM benchmarks, and permuted-anchor adapters. Results will be disaggregated by country and preference domain. A negative or null country is a result: it narrows the scope of the protocol rather than being silently pooled away.

Table 3: Predeclared main result shell. Values are filled only after the frozen multi-country experiment.

Country	Protocol arm	Synthetic diagnostics	External discrepancy	Interpretation
c_1	real anchor	[pending]	[pending]	[pending]
c_1	permuted-anchor placebo	[pending]	[pending]	anchor relevance test
c_1	country-prompt baseline	[pending]	[pending]	practical comparator
⋮	⋮	⋮	⋮	⋮

8. Discussion

The protocol changes the question posed to synthetic cultural agents. It does not ask whether an LLM can speak for a country, nor does it assume that country-level averages identify individual preferences. It asks whether a declared aggregate profile can be transformed into a reproducible policy whose synthetic provenance is visible and whose external relevance is testable.

This distinction has practical consequences. The method may be useful for exploratory measurement, hypothesis generation, survey-item design, and benchmark construction when its tests support those applications. It is not a substitute for collecting human data, and it should not be used to infer individual beliefs, to represent subgroups, or to support causal and structural claims without additional evidence. The correct scientific outcome is sometimes rejection: an anchor may fail to transmit, an adapter may fail to encode, a placebo may perform similarly, or an external criterion may not transport.

The protocol also makes clear why operator sensitivity is substantive. If a finding disappears when a scenario bank, selector, quality-control threshold, or training materialization changes, then the finding belongs to a fragile synthetic pipeline rather than to the aggregate anchor. Conversely, stability across those choices does not prove representation, but it increases confidence that the observed policy behavior is an attribute of the specified construction rather than an incidental artifact.

Code and data availability

The SCA2 implementation, frozen evaluation artifacts, and this working-paper source are available at `https://github.com/EconLLM-Lab/SCA2_PofW`. A claim-ready release will identify an immutable repository tag/commit, code license, model/API versions, and an artifact manifest. GPS data are documented by Falk et al. [4]; WVS Wave 7 data and documentation are available from the World Values Survey Association under DOI `https://doi.org/10.14281/18241.18` [8]. Raw survey microdata are not redistributed with the repository.

A. USA/Mexico pilot: a proof-of-concept diagnostic, not evidence of specificity

The existing two-country pilot is retained because it clarifies a design boundary. Under the primary unconditioned prompt, the same model emits identical option probabilities when scored against USA and Mexico; only the external target distribution differs. Thus a matched-versus-cross comparison measures which adapter's *fixed* output distribution lies closer to each population's observed responses. It does not show that the adapter knows or conditions on its target country at inference.

In the committed pilot, the Mexico adapter has mean TVD 0.4882 on Mexico versus 0.5211 for the base model, while the USA adapter has 0.6377 on Mexico. The corresponding mean matched-minus-cross TVD difference is 0.1495 with a question-resampling 95% interval $[0.0964, 0.2152]$ across 35 items. On USA outcomes, however, the Mexico adapter is approximately neutral relative to base (0.3888 versus 0.3958), while the USA adapter degrades (0.5258). These values are verified transcriptions of the committed evaluation summaries.

The pilot therefore does not establish country specificity. A single fixed adapter can dominate another fixed adapter against either country, and $C = 2$ cannot reveal a stable country-by-policy cross-over pattern. The item bootstrap quantifies sensitivity to this 35-item battery, not uncertainty from training runs, model generation, respondents, or country sampling. The pilot motivates—rather than substitutes for—the multi-country placebo and operator-sensitivity design in the main paper.

B. Adversarial scientific-design audit of the synthetic-generation pipeline

Scope and evidence. This independent appendix is a static audit of the specified pipeline and its tests, not an empirical validation of generated records. The targeted test suite was attempted but could not be collected in the review environments because `pandas`/`numpy` dependencies were unavailable. The inspected tests predominantly mock model completions (for example, `synthetic_generation/tests/test_generate.py:139-184` and `test_score.py:13-182`); no test-pass, psychometric-validity, inter-rater-reliability, or model-stability claim follows from this audit.

B.1 Blocker findings

Replace the LLM selector with a deterministic sign rule for non-zero target scores. The selector is explicitly instructed that positive z implies A and negative z implies B, while fixed A/B responses were already generated as positive/negative target loadings (`generate.py:170-180, 237-279`). It is therefore a stochastic, prompt-mediated implementation of a supplied deterministic label rather than an independent estimator of country-specific preference. It introduces label noise, permits irrelevant use of the other five profile dimensions, and depends on mutable model behavior. The primary construction should use A when $z > 0$ and B when $z < 0$, with $|z|$ near zero excluded, flagged, or handled under a separately justified probabilistic rule. The current exact-zero fallback is internally inconsistent: selector guidance asks for the "less extreme" option whereas QC treats zero as positive (`generate.py:250-255; score.py:204-206`).

Treat LLM scoring/QC as process diagnostics, not a validity screen. The scorer receives the target dimension and rubric, scores the already selected chosen/rejected texts, and QC retains rows whose scorer-derived target difference agrees with the known GPS sign (`score.py:46-64, 119-130, 204-235`). This is a second model confirming a target specified by design, not an independent measurement of cultural preferences or construct validity. No calibration against blinded human annotations, reliability analysis, held-out behavior, or threshold justification is present in the inspected code/tests. The scores may be useful operational diagnostics, but should not be described as validating a population-level construct.

B.2 Major findings

Selection-on-the-outcome and unfiltered contamination threaten validity. Scoring follows country-specific selection, so fixed A/B pairs are repeatedly scored in country-oriented order rather than once blind to country and choice (`generate.py:429-474; score.py:119-146`). Cross-dimension movement is calculated but does not determine `qc_status`; export removes only non-pass statuses, so high-contamination pairs can remain (`score.py:209-221; export.py:323-329`). The recommended design scores canonical A/B options once under blinded target and option order, orients scores arithmetically after deterministic labeling, reports all cross-loading distributions, and either predeclares a contamination gate or treats contamination as descriptive only.

Prompts attempt control but do not demonstrate it. The pair generator is instructed to vary only the target trait while holding the other five constant (`generate.py:175-180`); this is an instruction, not evidence of compliance. Optional curated anchors are injected into scenario and triplet prompts (`generate.py:72-76, 156-160`), creating a provenance channel that must be versioned and separated from evaluation materials. Country names are absent from profile text (`profiles.py:44-65`), a useful safeguard, but all six numeric dispositions remain available to selection (`generate.py:263-275`).

Generation reproducibility is incomplete. The manifest records config snapshots, GPS vectors, costs, scenario counts, timestamp, and a short Git hash; export shuffling uses a seed (`export.py:41-50, 367-399`). However, the pipeline seed is not passed to model calls and teacher/generator temperatures are non-zero (`config.py:225-228; generate.py:41-49, 194-202`). Manifests lack input hashes, resolved model revisions, dependency locks, raw-checkpoint hashes, prompts, and repository dirty state. Resume can reload current input data before rescoring an old checkpoint (`run.py:363-418`). Outputs should therefore be treated as dated stochastic realizations until immutable input and full-run provenance are archived.

B.3 Efficiency-preserving changes and a minimal ablation

Deterministic signs eliminate all per-country selector calls. The default 120 fixed triplets (six dimensions times 20 scenarios) otherwise generate 120 times the number of countries selection calls. Score each canonical pair once rather than once per country, orient scores locally by country sign, and compute QC arithmetic locally; this reduces country-linear scorer calls and removes country/choice-order leakage. Facets and scenarios are already batched, but per-triplet selector reasoning payloads are not used for labeling or QC and can be omitted or sampled for audit, while preserving raw responses and request metadata for remaining calls.

The minimum ablation uses a fixed scenario/A–B bank and compares: (i) deterministic signs versus legacy LLM selection; (ii) no QC versus scorer diagnostic-only versus scorer plus a predeclared contamination exclusion; and (iii) anchors off versus on. Each condition should retain at least three independently generated banks, stratified by dimension and z-score magnitude. Report label disagreement with the deterministic rule, QC retention, cross-loading distributions, blinded human ratings of target loading/non-target leakage/plausibility/option balance, and the full range of downstream conclusions. If legacy-selector and deterministic-sign estimates differ materially, the former is an unvalidated alternative rather than the primary dataset.

C. Protocol artifacts required for a claim-ready version

A claim-ready version will archive: (i) country inclusion rules; (ii) anchor and item-map versions; (iii) prompts, temperatures, model IDs, and endpoint versions; (iv) raw scenario, pair, selection, and scoring outputs; (v) QC tables; (vi) training manifests and adapter hashes; (vii) evaluation outputs; and (viii) analysis code for item, country, generation, and training-seed uncertainty. A remote API seed is not itself a sufficient reproducibility guarantee; materialized datasets are the primary artifacts.

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